

HADI KHAN

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CAREER SUMMARY

Computer Science undergraduate with a strong focus on the intersection of artificial intelligence, applied mathematics, and embedded systems. My work centres on developing mathematically grounded approaches to machine learning, with particular interest in Physics-Informed Neural Networks and their application to modelling physical phenomena such as signal propagation and system dynamics. Alongside my academic studies, I actively engage in electronics and low-level system design, including microcontroller configuration and PCB development, to bridge theoretical models with real-world implementations.

PROFESSIONAL EXPERIENCE

LAB DEMONSTRATOR | Manchester Metropolitan University

October 2026 – May 2027

Appointed as Lab Demonstrator (starting Oct 2026) for second-year Computer Science students. In this role, I will coordinate lab sessions to help students understand the content intuitively. This role will help me gain confidence and reinforce previous year topics for personal progression.

EDUCATION

BSC COMPUTER SCIENCE | Manchester Metropolitan University | UK, Manchester

2024 – 2027

Year 2 GPA: 81%

Relevant modules: Algorithms & Data Structures (81%), Computer Networks (80%), Operating Systems (predicted 88%), Ethical hacking (predicted 86%), Software Development Processes (predicted 75%)

AI SUMMER CAMP | Howest University of Applied Sciences | Belgium, Brugge

2026 – 2026

Attending a 5-day AI-oriented summer camp

TECHNICAL SKILLS

Programming: Python (advanced), C/C++ (intermediate), C#/Java (intermediate), MATLAB, LaTeX, Arduino

Software: CAD (Fusion 365, KiCad), Overleaf, Simulink

PROJECTS & PUBLICATIONS

- Developed a gradient descent model in Python using pure mathematical calculations. Developed optimisation algorithms, such as momentum and ADAM, to converge more complex functions. Visualised this using matplotlib's animation package, and documented algorithms on a LaTeX board.
- Built a simulation of a wide area network using Python, which uses Dijkstra's and Brandes' algorithm to gather metrics, which are used to train a Random Forest Regression model to predict weak nodes and edges in a network. Developed a WAN in Cisco Packet Tracer as a prototype for the Python script.

REFERENCES

Nayab Rana | Contact: nayab.rana@barclays.com

JavaScript Engineer – Barclays Bank

Anthony Kleerekoper | Contact: A.Kleerekoper@mmu.ac.uk

Senior Lecturer in Computer Science – Manchester Metropolitan University

Michael Jones | Contact: michael.jones@mmu.ac.uk

Programme Leader BSc (Hons) Computer Science – Manchester Metropolitan University